

Sufficient-Statistic Formula for Optimal Stimulus Spending

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Implicit formula for optimal stimulus spending:

$$\frac{g/c - g/c^*}{g/c^*} = 2\varepsilon \times m \times \underbrace{\left(\frac{u - u^*}{u^*} \right)}_{\text{function of } g/c}$$

→ We want an explicit formula for optimal stimulus spending → formula involving initial unemployment gap $\frac{u_0 - u^*}{u^*}$ and other sufficient statistics.

Q. we are at $\frac{u_0 - u^*}{u^*}$ & spending is g/c^* .

how much $\uparrow$ recession should spending increase/decrease?

To make formula explicit, we express $\frac{u - u^*}{u^*}$ as a function of $\frac{u_0 - u^*}{u^*}$ & $\frac{g/c - g/c^*}{g/c^*}$.

First-order approximation of $\frac{u - u^*}{u^*}$ around initial situation

$$\frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} + \frac{1}{u^*} \times \underbrace{\frac{du \times g/c^*}{dg/c}}_{\frac{du}{d \ln g/c}} \left[\frac{g/c - g/c^*}{g/c^*} \right]$$

$$\frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} + \frac{1}{u^*} \times \frac{du}{d \ln g/c} \times \left[\frac{g/c - g/c^*}{g/c^*} \right]$$

need to rewrite

Compute $du/d \ln g/c$: evaluated at $g/c^*, u^*$

$$\frac{du}{d \ln g/c} = \frac{du}{dg} \times \frac{dg}{d \ln g/c} = -m \times \frac{dg}{d \ln g/c}$$

Compute $dg/d \ln g/c$

$$c = 1 - (u + v) - g$$

$$\frac{dc}{dg} = \frac{-d(u+v)}{dg} = -1$$

at $u^*, g/c^*$

$$\frac{d(u+v)}{dg} \text{ at } u^* = 0$$

$\hookrightarrow \frac{d(u+v)}{du} \times \frac{du}{dg}$
 $0 \rightarrow \frac{du}{dg}$

$$\frac{dc}{dg} = -1 \quad \text{at } u^* \quad \text{at } g/c^* \quad -1$$

$$\frac{d \ln g/c}{dg} = \frac{1}{g/c} \times \frac{dg/c}{dg} = \frac{1}{g/c} \times \left[\frac{1}{c} - \frac{g/c}{c^2} \times \frac{dc}{dg} \right]$$

$$\frac{d \ln g/c}{dg} = \frac{c^*}{g^*} \times \left[\frac{1}{c^*} + \frac{g^*}{c^{*2}} \right] = \frac{1}{g^*} + \frac{1}{c^*}$$

$$\Rightarrow \frac{du}{d \ln g/c} = -m \times \left(\frac{1}{\frac{1}{g^*} + \frac{1}{c^*}} \right) = -\frac{m}{2} \times \left[\frac{2}{\frac{1}{g^*} + \frac{1}{c^*}} \right]$$

harmonic mean of g^*, c^*

$$\Rightarrow \frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} - \frac{m}{2 u^*} \left(\frac{2}{\frac{1}{g^*} + \frac{1}{c^*}} \right) \left(\frac{g/c - g/c^*}{g/c^*} \right)$$

Initial unemployment g/c
suff. stat.
stimulus spending

$\Rightarrow$ plug this into implicit formula to make it explicit.

$$\frac{g/c - g/c^*}{g/c^*} = 2 \varepsilon m \times \left(\frac{u_0 - u^*}{u^*} \right) - \underbrace{\varepsilon m^2 \times \frac{1}{u^*} \times \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}}}_{= Z} \times \frac{g/c - g/c^*}{g/c^*}$$

$$\left[1 + \varepsilon m^2 Z \right] \times \frac{g/c - g/c^*}{g/c^*} = 2 \varepsilon m \left(\frac{u_0 - u^*}{u^*} \right)$$

Explicit formula for optimal stimulus spending

$$\frac{g/c - g/c^*}{g/c^*} = \frac{2 \varepsilon m}{1 + z \varepsilon m^2} > \frac{u_0 - u^*}{u^*}$$

stimulus spending

init. of unemployment gap

ε = elasticity of substitution b/w g & c

m = unemployment multiplier

$$z = \frac{1}{u^*} \times \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}}$$